

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	13 July 2026
Team ID	SWTID-2026-7493
Project Name	BookStore
Maximum Marks	4 Marks

Technical Architecture:

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Provides web interface for Users, Sellers, and Admin	React.js, Bootstrap
2	Application Logic – Authentication	Handles user registration, login, and JWT authentication	Node.js, Express.js, JWT, bcrypt
3	Application Logic – Book Management	Handles CRUD operations for books	Express.js, Mongoose
4	Application Logic – Order Management	Processes book orders and order history	Express.js
5	Database	Stores users, books, and orders	MongoDB
6	API Communication	Connects frontend and backend	REST API, Axios
7	File Upload	Uploads book cover images	Multer
8	Version Control	Source code management	Git & GitHub

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used in the project	React.js, Express.js, Node.js
2	Security Implementations	Secure login and role-based access	JWT, bcrypt
3	Scalable Architecture	Three-tier architecture	React + Express + MongoDB
4	Availability	Local server-based application	Node.js Server
5	Database	Persistent storage	MongoDB